

MEDIATEK

MT8676 Android IPO User Manual

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Version History

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1 IPO Overview

1.1 Base Introduction

This document primarily introduces the MT8676 IPO quick boot function and provides analysis methods for common IPO issues. IPO, short for Instant Power On, refers to the quick boot feature, which is very practical for quickly accessing the reverse camera view and activating the car's recording functions after the car's ACC is turned on.

1.2 Why IPO is Needed

- In the current smart automotive control systems, the startup time of devices is an important performance metric.
- IPO is a software solution that allows devices to boot within a configurable time without additional hardware costs. It utilizes the Android suspend function to create a new user experience.
- During the IPO phase, most processes and services running in the system are stopped or killed by the ActivityManagerService to prevent unintended behavior. Once the system enters the suspend state, users can wake up the system through external or remote events to exit IPO mode, thus achieving rapid startup.
- Users may ignite and turn off their vehicles daily and have expectations for the speed of switching; they do not want to wait too long.
- Even if users feel that the car system has been turned off, it is only a pseudo shutdown. In reality, the system has not truly shut down but has entered an early_suspend or suspend state, and the car system can still receive remote commands.

2 Abbreviations

Table 2-1. Abbreviations

Abbreviation	Explanation
HU	Head Unit
IHUUH	Infotainment Head Unit
IPO	Instant Power On
IVI	In-Vehicle Infotainment
T-Box	Telematics-BOX

3 Architectural Overview

3.1 Android IPO Architecture

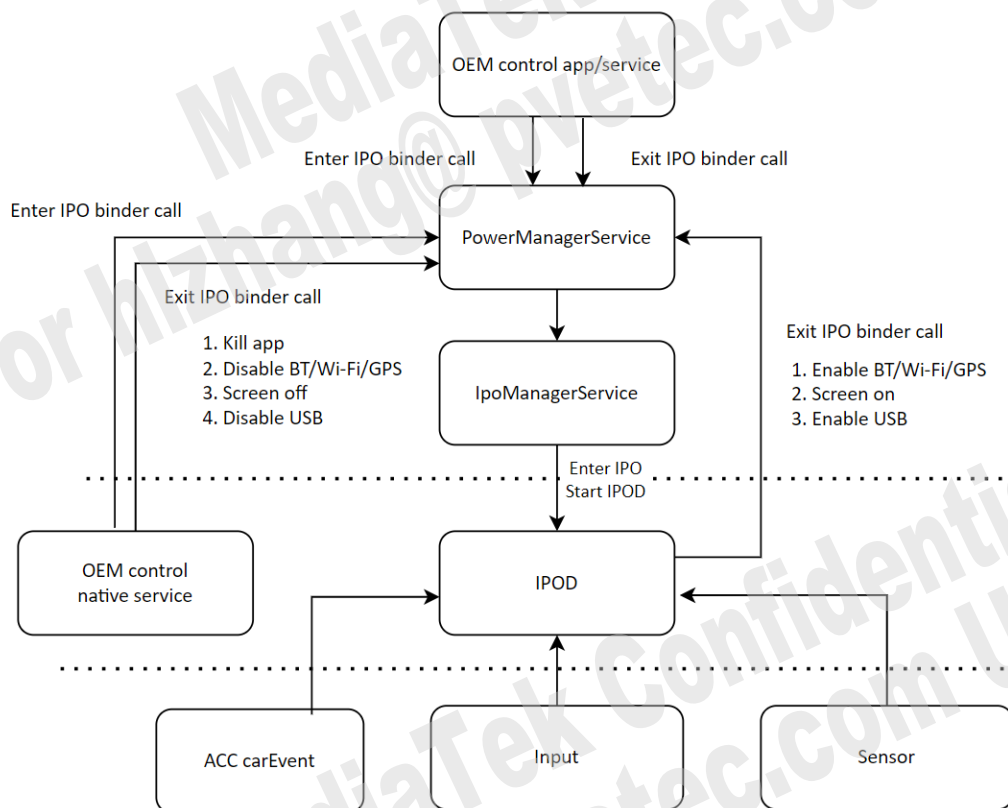


Figure 3-1. IPO architecture

3.2 Two Scenarios of Hardware Power Management

1. Independent power management logic, not dependent on the Kernel Suspend process (such as Camera/Display, etc.)
The driver powers down the device when it detects no access and powers up and acquires a WakeLock when accessed.
2. Dependent on the Kernel Suspend/Resume architecture
 - Hardware Driver implements the Suspend/Resume functions required by the Kernel Driver.
 - If no one holds a WakeLock, the system will call the `Suspend()` function of each HW Driver when entering the Suspend state.
 - When the system is awakened and enters the Resume state, it will also call the `Resume()` function of each HW Driver.
 - HW Drivers independently implement the Suspend/Resume functions to control powering up and down on their own.

The power management schemes corresponding to the above two scenarios are as follows:

1. During IPO shutdown, trigger all users to relinquish access to hardware devices: First stage power down.
2. During IPO shutdown, clear hardware usage scenarios, release WakeLock, and enter suspend state: Second stage power down.

3.3 Power Management Schemes for Each Hardware During Entry and Exit IPO

Table 3-1. Power management schemes for each hardware during entry and exit IPO

Module	Suspend Stage	IPO Shutdown Handling	IPO Startup Handling
GPS	2	1. IPO kills HU-side AP to clear GPS application scenario 2. Stop using GPS; GPS SW enters Suspend	No handling during IPO startup
BT	1	IPO will turn off BT; BT SW enters Suspend	IPO will turn on BT
WIFI	1	IPO will turn off WIFI; WIFI SW enters Suspend	IPO will turn on WIFI
TTY	2	1. Clear TTY usage scenario 2. Kernel Suspend -> TTY Driver Suspend	No handling during IPO startup Kernel Resume -> TTY Driver Resume
SPI	2	1. Clear SPI usage scenario 2. Kernel Suspend -> SPI Driver Suspend	No handling during IPO startup Kernel Resume -> Corresponding Driver Resume
Camera	1	1. IPO kills HU-side AP to clear Camera application scenario 2. EVS stops Camera usage upon ACC OFF 3. Camera Driver powers down	No handling during IPO startup If Camera is in use, Camera Driver powers up
Sensor	2	1. IPO kills HU-side AP to clear sensor application scenario 2. Check for remaining sensor usage scenarios 3. Kernel Suspend -> Sensor Driver Suspend	No handling during IPO startup Kernel Resume -> Corresponding Driver Resume
EPROM	2	1. IPO kills HU-side AP to close EPROM application scenario 2. Check for other usage scenarios 3. Kernel Suspend -> Sensor Driver Suspend	No handling during IPO startup Kernel Resume -> Corresponding Driver Resume
Display	1	1. IPO calls PMS_FW to enter Early Suspend 2. Display Driver powers off Display	1. IPO startup calls PMS_FW to switch backlight 2. Display Driver notifies MCU via GPIO
Mic noise elimination	2	1. Enter IPO shutdown and remove usage scenario 2. Driver enters Suspend and powers down	No handling during IPO startup Kernel Resume -> Corresponding Driver Resume
Mic D/A conversion	2	1. Enter IPO shutdown, remove usage scenario 2. Driver enters Suspend and powers down	No handling during IPO startup Kernel Resume -> Corresponding Driver Resume
USB Hub	2	1. Enter IPO shutdown and remove usage scenario 2. Driver enters Suspend and powers down	No handling during IPO startup Kernel Resume -> Corresponding Driver Resume
Ethernet	2	1. Enter IPO shutdown and remove usage scenario	No handling during IPO startup

Module	Suspend Stage	IPO Shutdown Handling	IPO Startup Handling
		2. Driver enters Suspend and powers down	Kernel Resume -> Corresponding Driver Resume

3.3.1 Status of CPU and Processes in Different Modes

- Software Process States: Running (occupying CPU for execution), Suspend (actively relinquishing CPU resources and waiting for execution), Killed (terminated))
- CPU States: Wakeup (screen on), Early Suspend (awake state & screen off), Suspend (sleep mode)

Table 3-2. Status of CPU and processes in different modes

	MT8676 Android			
	CPU	T-Box Process	Java Core Process /Native Process	Task Process
Full Work Mode	Early Suspend/ Suspend/Wakeup	Suspend/Running IPO Pseudo Shutdown State	Suspend/Running	Suspend/Running
Partial Work Mode	Early Suspend	Running	Running	Killed
Sleep mode	Suspend	Suspend	Suspend	Killed

3.3.2 IPO Interface Description

Table 3-3. IPO interface description

Interface/Structure	Description
public static void lowpowerStartIPO()	PowerManagerService provided IPO shutdown interface: 1. Kill apps not on the whitelist, turn off peripherals such as BT/WIFI/GPS, and turn off the screen. 2. AP releases the WakeLock, and the system enters sleep mode.
public static void lowpowerStopIPO(int flag)	PowerManagerService IPO startup exit interface Parameter: 0 (screen off), 1 (screen on) The system enters Full Work mode.

3.3.3 IPO Interface Invocation Timing

3.3.3.1 Entering the IPO Shutdown Process

- Red box: Apps not on the whitelist can receive the ACTION_SHUTDOWN_HU broadcast and perform cleanup actions.
- Purple box: Call the Android layer interface to turn off peripherals and remove peripheral usage scenarios.
- Yellow box: Kill apps not on the whitelist and native processes on the blacklist.

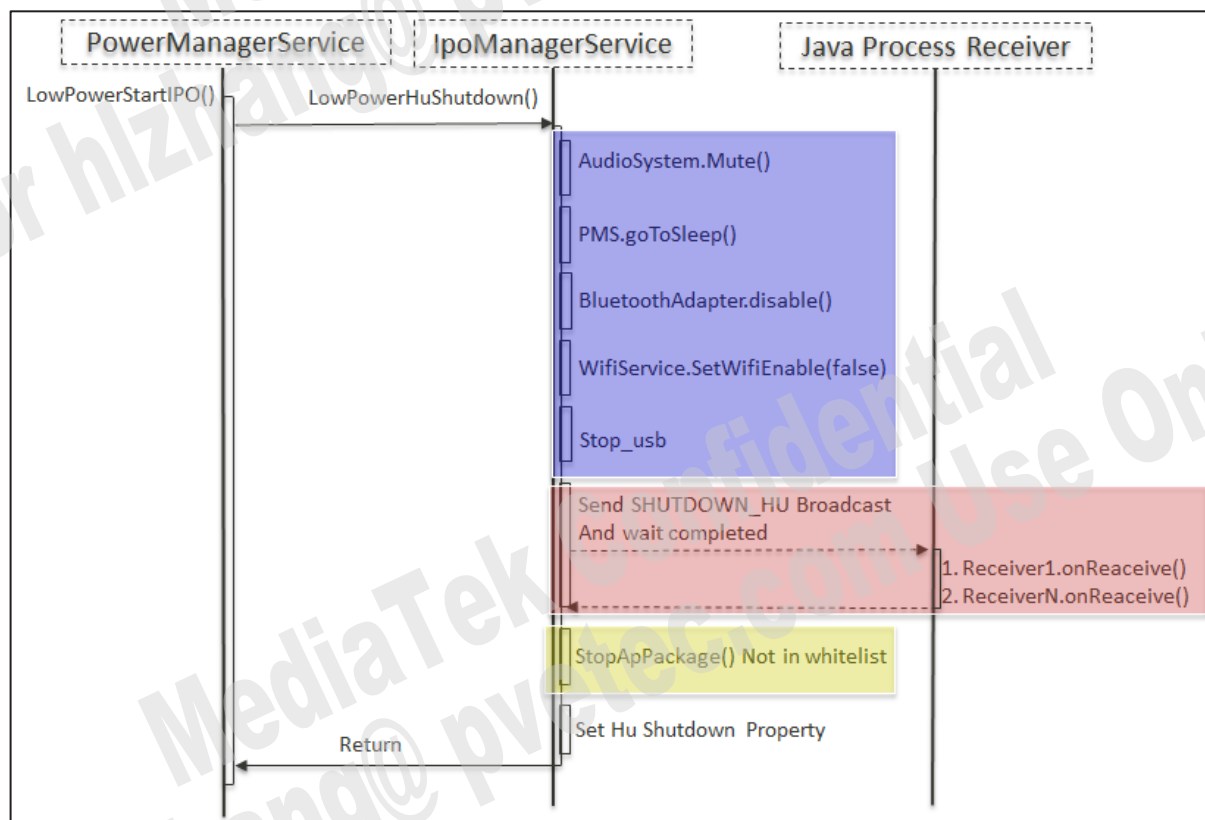


Figure 3-2. IPO shutdown process

3.3.3.2 Exiting the IPO Startup Process

Ignition wakeup interrupt/Modem call or SMS -> Wakeup Kernel -> Execute IPO exit.

Boot animation and boot process are handled in parallel.

- Purple box: Actively wake up devices that were turned off in the first stage before broadcasting.
- Red box: Can receive the ACTION_BOOT_HU broadcast to perform initialization actions.
- Yellow box: Send the native Android boot broadcast to start apps expected to auto-start on boot.

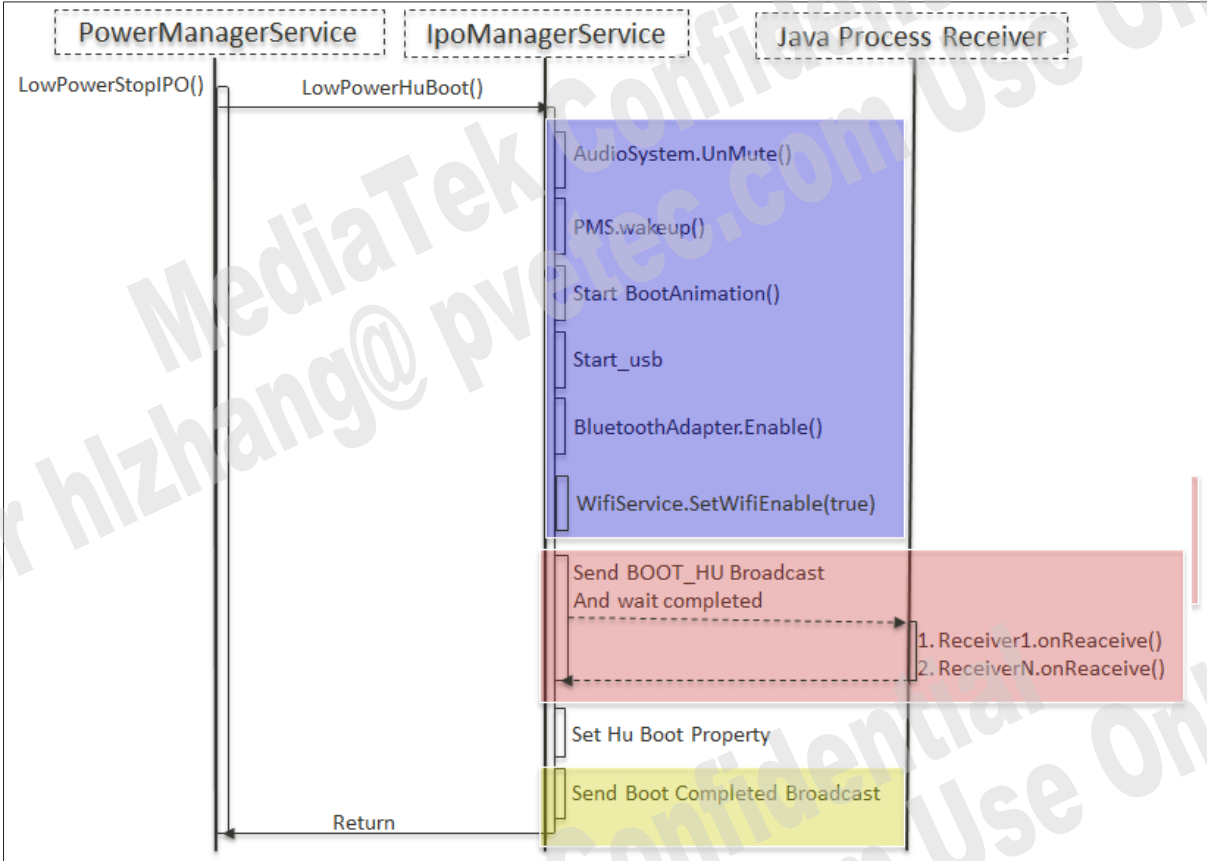


Figure 3-3. Process of exiting the IPO startup

4 Configuration/Customization Guide

4.1 Modifications on the HU Side

Customized apps can listen to the following two broadcasts:

- `com.mediatek.intent.action.ACTION_SHUTDOWN_HU`: Perform some actions before shutdown
- `com.mediatek.intent.action.ACTION_BOOT_HU`: Perform some actions before startup

4.2 Introduction to Code Modifications

Table 4-1. Modifications related to IPO code

AOSP Repo	Modified Files	Reason for Modification
frameworks/base	core/java/android/os/IPowerManager.aidl	Added interface to enter IPO Service.
frameworks/base	core/res/AndroidManifest.xml	Defined broadcasts for IPO startup and shutdown.
frameworks/base	services/core/java/com/android/server/power/PowerManagerService.java	Added interface to enter IPO Service.
frameworks/base	services/core/Android.bp	Load MediaTek IPO Jar file.
frameworks/base	services/core/java/com/android/server/policy/PhoneWindowManager.java	During IPO shutdown, skip key events.
vendor/mediatek/proprietary/frameworks/opt/ipod	java/com/mediatek/ipomanager/IpoManagerService.java	IPO core class.
vendor/mediatek/proprietary/external/ipod	*	After entering IPO, IPOD exits IPO upon detecting a power key event.

4.3 USB Connection

After entering IPO, it is necessary to switch to Device mode to disconnect USB and enable sleep; upon exiting IPO, switch back to Host mode to prevent the USB kernel from holding a lock, which could prevent entering Suspend mode. Upon exiting IPO, restore the USB connection.

Code path: `vendor/mediatek/proprietary/external/ipod/ipod.rc`

```

18 on property:sys.ipo.disable.usb=1
19     write /sys/class/udc/musb-hdrc/device/cmode 0
20
21 on property:sys.ipo.disable.usb=0
22     write /sys/class/udc/musb-hdrc/device/cmode 1

```

Figure 4-1. USB mode switching

4.4 How to Customize the Animation for Entering and Exiting the IPO Process

Code path: vendor/mediatek/proprietary/frameworks/opt/ipo/IpoManagerService.java

```

304     private IpoManagerService(Context context) {
305         myLog( "start IpoManagerService");
306         if(SystemProperties.get("persist.sys.ipo.debug", "").equals("1")){
307             localLOGV = true;
308         }
309         mWelcomPackage = SystemProperties.get("ro.odm.welcomeactivity.package", "com.incall.welscreen");
310         mWelcomClass = SystemProperties.get("ro.odm.welcomeactivity.class", "com.incall.activity.WelcomeActivity");
311         mBootAnimService = SystemProperties.get("ro.odm.bootanimation.name", "bootanim");
312         mShutAnimService = SystemProperties.get("ro.odm.shutdownanimation.name", "bootanim");
313         mBootAnimTime = Integer.valueOf(SystemProperties.get("ro.odm.bootanimation.time", "2"));
314         mShutAnimTime = Integer.valueOf(SystemProperties.get("ro.odm.shutdownanimation.time", "2"));
315         MAX_BOOT_TIME = mBootAnimTime*1000+1000;
316         MAX_SHUTDOWN_TIME = mShutAnimTime*1000+1000;

```

Figure 4-2. Animation customization

The default boot animation time is 2 seconds, which can be customized according to requirements.

4.5 Custom IPO Whitelist

Processes on the whitelist will not be killed when entering IPO.

Code path: vendor/mediatek/proprietary/frameworks/opt/ipo/IpoManagerService.java

- To set the whitelist using hard coding, please refer to the following diagram:

```

215     private static final String[] mHardCodeShutdownList = {
216         "system",
217         "com.android.bluetooth",
218         "com.android.phone",
219         "android.process.acore",
220         "com.android.systemui",
221         "com.android.car",
222         "com.android.car.carlauncher",
223         "com.android.providers.media.module",
224     };

```

Figure 4-3. Set the whitelist using hard coding

- To set the whitelist using Dynamic method:
setprop persist.ipo.shutdown.process.wl

Example: /com.xx.yy/com.zz.ww. Please refer to [Figure 4-4](#).

```

245 // Customization +
246 String processList = SystemProperties.get("persist.ipo.shutdown.process.wl", null);
247 if ( processList != null ) {
248     myLog("Process whitelist: " + processList);
249     ArrayList<String> processArrayList = new ArrayList<String>();
250     parseStringIntoArray("/", processList, processArrayList);
251
252     for (int i = 0; i < processArrayList.size(); ++i) {
253         String item = processArrayList.get(i);
254         if (item.length() > 0) {
255             if (item.startsWith("!") && item.length() > 1
256                 && sShutdownWhitelist.contains(item.substring(1))) {
257                 sShutdownWhitelist.remove(item.substring(1));
258             } else if (!item.startsWith("!") && !item.matches("^\\s*$")
259                 && !sShutdownWhitelist.contains(item)) {
260                 sShutdownWhitelist.add(item);
261             }
262         }
263     }
264 }
265 // Customization -

```

Figure 4-4. Whitelist code

4.6 Killing Processes When Exiting IPO

Configure `persist.ipo.prebootkill.list`.

4.7 Notifying Java Processes When Entering/Exiting IPO

When IPO is shutting down, receive the broadcast `com.mediatek.intent.action.ACTION_SHUTDOWN_HU`.

When IPO is starting up, receive the broadcast `com.mediatek.intent.action.ACTION_BOOT_HU`.

4.8 Notifying Native Processes When Entering/Exiting IPO

Add property trigger into `init.rc` file.

Sample code:

```

12 on property:sys.hu.status=shutdown-done
13     start vendor.ipod
14
15 on property:sys.hu.status=boot-done
16     stop vendor.ipod

```

Figure 4-5. Notify native processes using property triggers in `init.rc` file

5 Frequently Asked Questions

5.1 Key Logs Generated When Shutting Down IPO

Key Word: LowPowerStartIPO/ IpoManagerService/ IPODMAIN/ipod

```
03-07 08:30:57.955 1188 1320 D PowerManagerService: LowPowerStartIPO //PMS IPO
entrance
03-07 08:30:57.956 1188 1320 D CarTbox : [IpoManagerService] Power_status_2.2
//IpoManagerService entrance
03-07 08:30:57.956 1188 1320 D CarTbox : [IpoManagerService] lowPowerHuShutdown start
03-07 08:30:57.957 1188 1401 D CarTbox : [IpoManagerService]handleMessage msg.what =
SHUT_ANIMATION_START
03-07 08:30:57.957 1188 1401 D CarTbox : [IpoManagerService] startShutAnimation
03-07 08:30:57.966 1188 1401 D CarTbox : [IpoManagerService]handleMessage msg.what = AUDIO_STOP
03-07 08:30:57.966 1188 1401 D CarTbox : [IpoManagerService]muteSystem
03-07 08:30:57.967 1188 1401 D CarTbox : [IpoManagerService]handleMessage msg.what = BT_STOP
03-07 08:30:57.968 1188 1401 D CarTbox : [IpoManagerService]stop_bt
...
03-07 08:30:59.158 1188 1401 D CarTbox : [IpoManagerService]handleMessage msg.what = APP_STOP
03-07 08:30:59.161 1188 1401 D CarTbox : [IpoManagerService]forceStopKillPackages //AMS kill
process
03-07 08:30:59.161 1188 1401 D CarTbox : [IpoManagerService]whitelist = system
03-07 08:30:59.161 1188 1401 D CarTbox : [IpoManagerService]whitelist = com.android.bluetooth
03-07 08:30:59.161 1188 1401 D CarTbox : [IpoManagerService]whitelist = com.android.phone
03-07 08:30:59.162 1188 1401 D CarTbox : [IpoManagerService]whitelist = android.process.acore
03-07 08:30:59.162 1188 1401 D CarTbox : [IpoManagerService]whitelist = com.android.systemui
03-07 08:30:59.162 1188 1401 D CarTbox : [IpoManagerService]whitelist = com.android.car
03-07 08:30:59.162 1188 1401 D CarTbox : [IpoManagerService]whitelist = com.android.car.carlauncher
03-07 08:30:59.162 1188 1401 D CarTbox : [IpoManagerService]whitelist =
com.android.providers.media.module
...
03-07 08:31:00.459 1188 1401 D CarTbox : [IpoManagerService]lowPowerHuShutdown Done
03-07 08:31:00.460 1188 1320 D CarTbox : [IpoManagerService] Power_status_3
//IpoManagerService done

03-07 08:31:00.506 4783 4783 I IPODMAIN: Can't load library: dlopen failed: library
"/system/lib/libipod.so" not found //start ipod
```

5.2 Key Logs Generated When Enabling IPO Using Commands

Key Word: LowPowerStopIPO/ IpoManagerService

```

03-07 08:37:56.648 1188 1639 D PowerManagerService: LowPowerStopIPO
03-07 08:37:56.648 1188 1639 D CarTbox : [IpoManagerService] Power_status_6.2
03-07 08:37:56.649 1188 1639 D CarTbox : [IpoManagerService] lowPowerHuBoot start
03-07 08:37:56.649 1188 1639 D CarTbox : [IpoManagerService]huStatus:shutdown-done
03-07 08:37:56.649 1188 1639 D CarTbox : [IpoManagerService]pre requestdscreen:1
03-07 08:37:56.651 1188 1401 D CarTbox : [IpoManagerService]handleMessage msg.what = AUDIO_RESTORE
03-07 08:37:56.652 1188 1401 D CarTbox : [IpoManagerService]unmuteSystem
03-07 08:37:56.653 1188 1401 D CarTbox : [IpoManagerService]handleMessage msg.what =
BOOT_ANIMATION_START
03-07 08:37:56.653 1188 1401 D CarTbox : [IpoManagerService] startBootAnimation
03-07 08:37:56.664 1188 1401 D CarTbox : [IpoManagerService]handleMessage msg.what = WAIT_MESSAGE
...
03-07 08:37:57.175 1188 1401 D CarTbox : [IpoManagerService]handleMessage msg.what = USB_RESTORE
03-07 08:37:57.175 1188 1401 D CarTbox : [IpoManagerService]start_usb

```

5.3 Key Logs Generated When Enabling IPO Using the Power Button

Key Word: IPO_RESUME_REASON /LowPowerStopIPO/ IpoManagerService

```

03-07 08:45:14.497 4839 4846 I IPODMAIN: <ipo resume system>IPO_RESUME_REASON:powerkey.
03-07 08:45:14.498 4839 4846 I IPODMAIN: LowPowerStopIPO

```

5.4 Key Logs Generated When Enabling IPO Due to a Collision

Key Word: GSENSOR_COLLISION /LowPowerStopIPO

```

09-15 14:41:54.560 7545 7551 I IPODMAIN: handleEvent: event 2 mHasCallback = 0
09-15 14:41:54.560 7545 7551 I IPODMAIN: IPO_SHUTDOWN_PROP: shutdown-done
09-15 14:41:54.560 7545 7551 I IPODMAIN: sendBroadcastMessage android.intent.action.GSENSOR_COLLISION
09-15 14:41:54.560 7545 7551 I IPODMAIN: sendBroadcastMessage(): Action: c, Value: 2
09-15 14:41:54.603 7545 7551 I IPODMAIN: LowPowerStopIPO
09-15 14:41:54.603 1174 7404 D PowerManagerService: LowPowerStopIPO

```

5.5 How to Troubleshoot High Power Consumption After Entering IPO

1. Command: `dumpsys power`
Check the presence of `partial WakeLock`.
2. Check UART logs to identify which system component or process causes the system to wake up.
3. Command: `cat /sys/kernel/debug/wakeup_sources`
Check `Kernel wakeup_sources`.

5.6 IPO Debugging Commands

1. By default, the Kernel stops printing UART logs a few seconds after boot. To keep UART logs continuously printing (this will also take effect after a reboot), use the following command:
`adb shell setprop persist.vendor.uartconsole.enable 1`
2. Entering IPO will disconnect the USB adb connection. If you do not want it to disconnect, you can enter the following command after booting:

```
adb shell setprop sys.ipo.usb 1
```

3. Enter IPO command: `adb shell service call power 67`
4. Exit IPO command: `adb shell service call power 68 i32 1`

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